

roulette wheel could be exploited by a sufficiently savvy gambler. Even if we are creating a PRNG for a purpose that has nothing to do with roulette, we can ask ourselves, “If I use this PRNG to govern a roulette application, would I lose all my money?” This intuitive “roulette test” is a reasonable criterion for judging how good any PRNG is. Our LCG might pass the roulette test if we never do more than 32 spins, but after that, a gambler could notice the repeating pattern of outputs and start to place bets with perfect accuracy. The short period of our LCG has caused it to fail the roulette test.

Because of this, it helps to ensure that a PRNG has a long period. But in a case like a roulette wheel with only 32 slots, no deterministic algorithm can have a period longer than 32. That’s why we often judge a PRNG by whether it has a *full period* rather than a long period. Consider the PRNG that we get by generating `list_random(1, 2, 24)`:

[1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 1]

In this case, the period is 12, which may be long enough for very simple purposes, but it is not a full period because it does not encompass every possible value in its range. Once again, a savvy gambler might notice that even numbers are never chosen by the roulette wheel (not to mention the simple pattern the chosen odd numbers follow) and thereby increase their winnings at our expense.

Related to the idea of a long, full period is the idea of *uniform distribution*, by which we mean that each number within the PRNG's range has an equal likelihood of being output. If we run `list_random(1, 18, 36)`, we get:

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[1, 19, 1, 19, 1, 19, 1, 19, 1, 19, 1, 19, 1, 19, 1, 19, 1, 19, 1, 19, 1,  
19, 1, 19, 1, 19, 1, 19, 1, 19, 1, 19, 1]
```

Here, 1 and 19 each have a 50 percent likelihood of being output by the PRNG, while each other number has a likelihood of 0 percent. A roulette player would have a very easy time with this non-uniform PRNG. By contrast, in the case of `list_random(29, 23, 32)`, we find that every number has about a 3.1 percent likelihood of being output.